

“Is there a text in this class?”

Potential and limits of text-based empirical research
in innovation studies

Summer School

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A small, stylized pencil icon with an orange body and a blue eraser, positioned above the letter 'S' in the title.

IS THERE A TEXT IN THIS CLASS ?

The Authority of
Interpretive Communities

STANLEY FISH

Stanley Fish (1938-) Literary and legal theorist

Debate on the interpretation of literary texts

- is there a “right” interpretation of a literary text?
- who has the authority to establish the right interpretation?

Stanley Fish’ experiment

- ask a class of students to comment a text from a poem of an unknown author
- assign a homework

All students performed the analysis in a proper way and submitted the homework, but.....

Key points

The text supposes a reader

Texts are polysemic

Texts are ambiguous

Texts are heterogeneous

Texts are organized hierarchically

Texts are narratives

The text supposes a reader.

The interpretation of the text
comes from the interaction between the author
and the reader

The text, once produced, takes its own autonomy

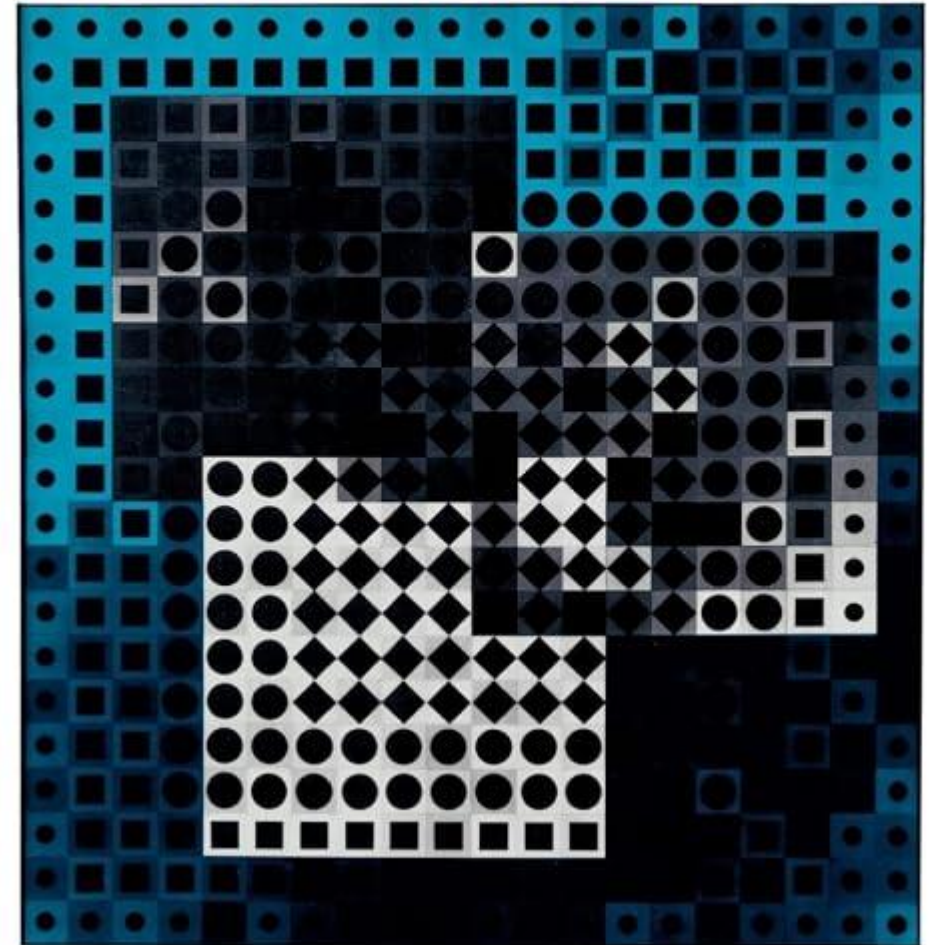
- it travels independently on the will of the author
- it interacts with the expectations of the reader

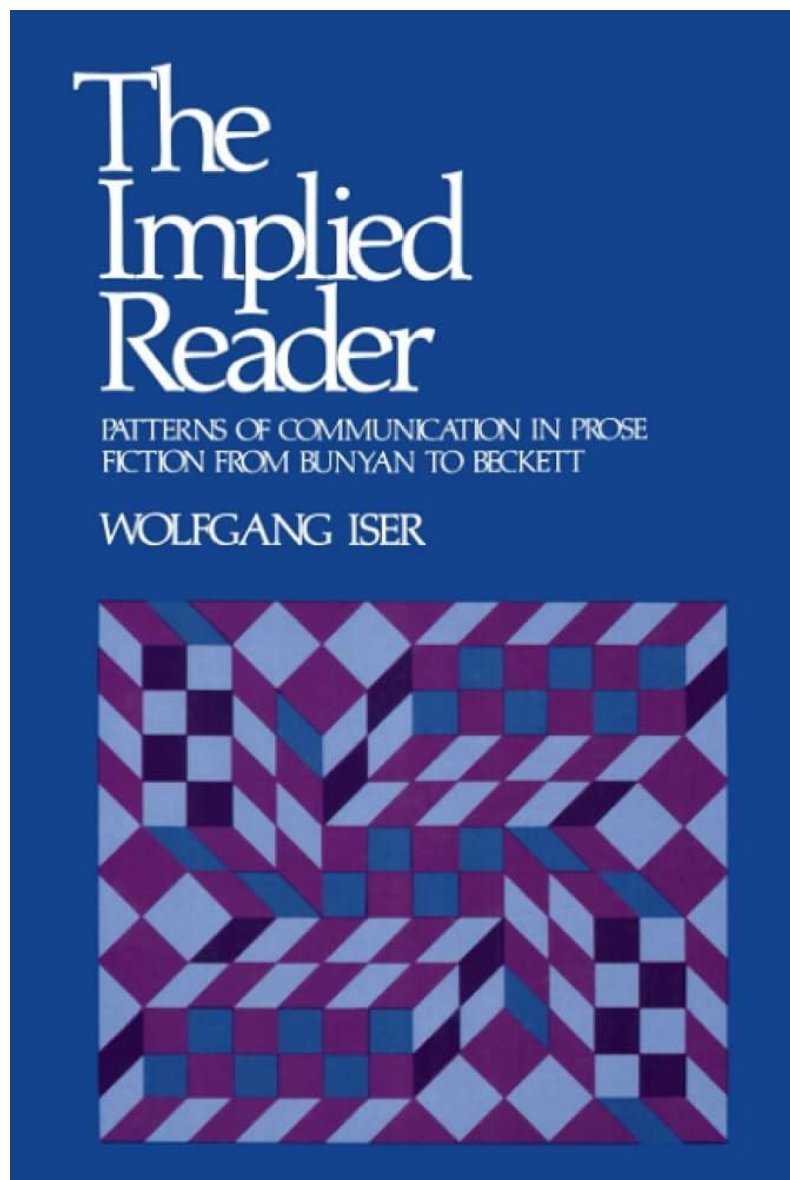
To interpret a text means to reconstruct the
“history of reception”
or the horizon of expectations of its readers
over time

WOLFGANG ISER

The Act of Reading

A THEORY OF AESTHETIC RESPONSE

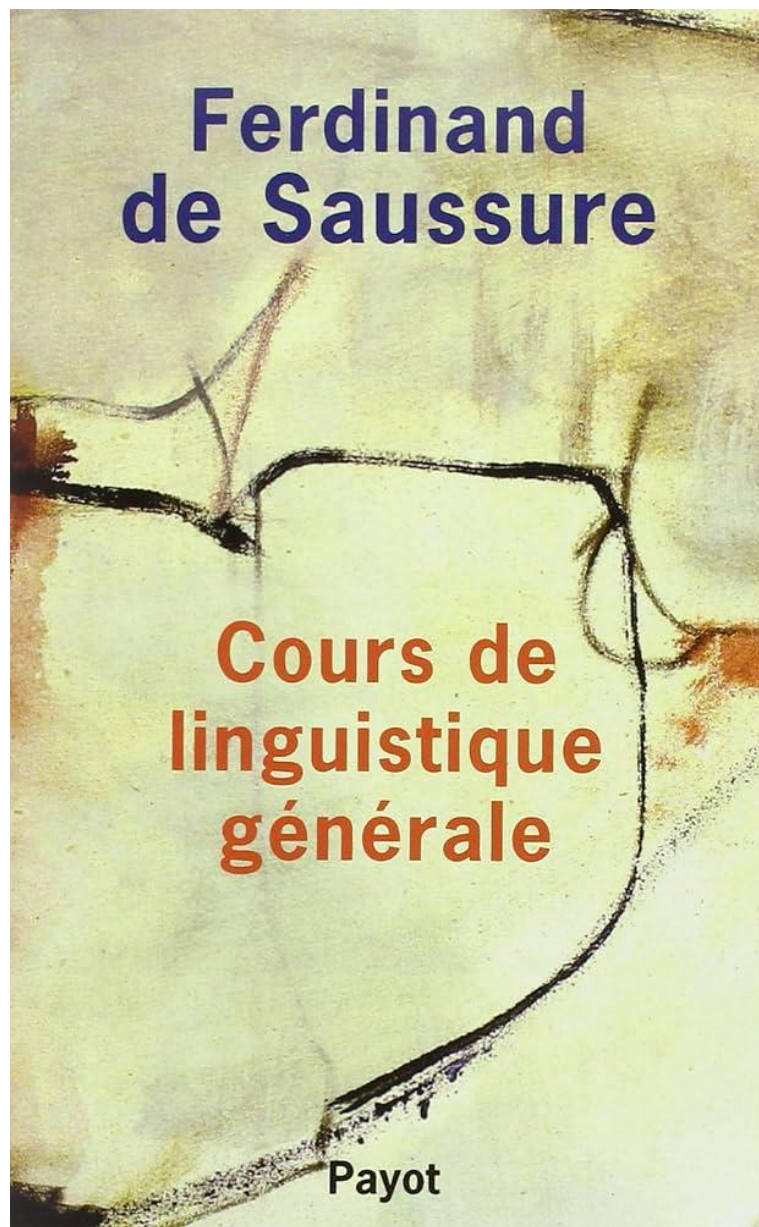




The author of a text has in mind a reader (“implied reader”) with whom he/she enters into dialogue.

But the reader may build up his/her experience of the text

Intentio auctoris vs Intentio operis (Umberto Eco)

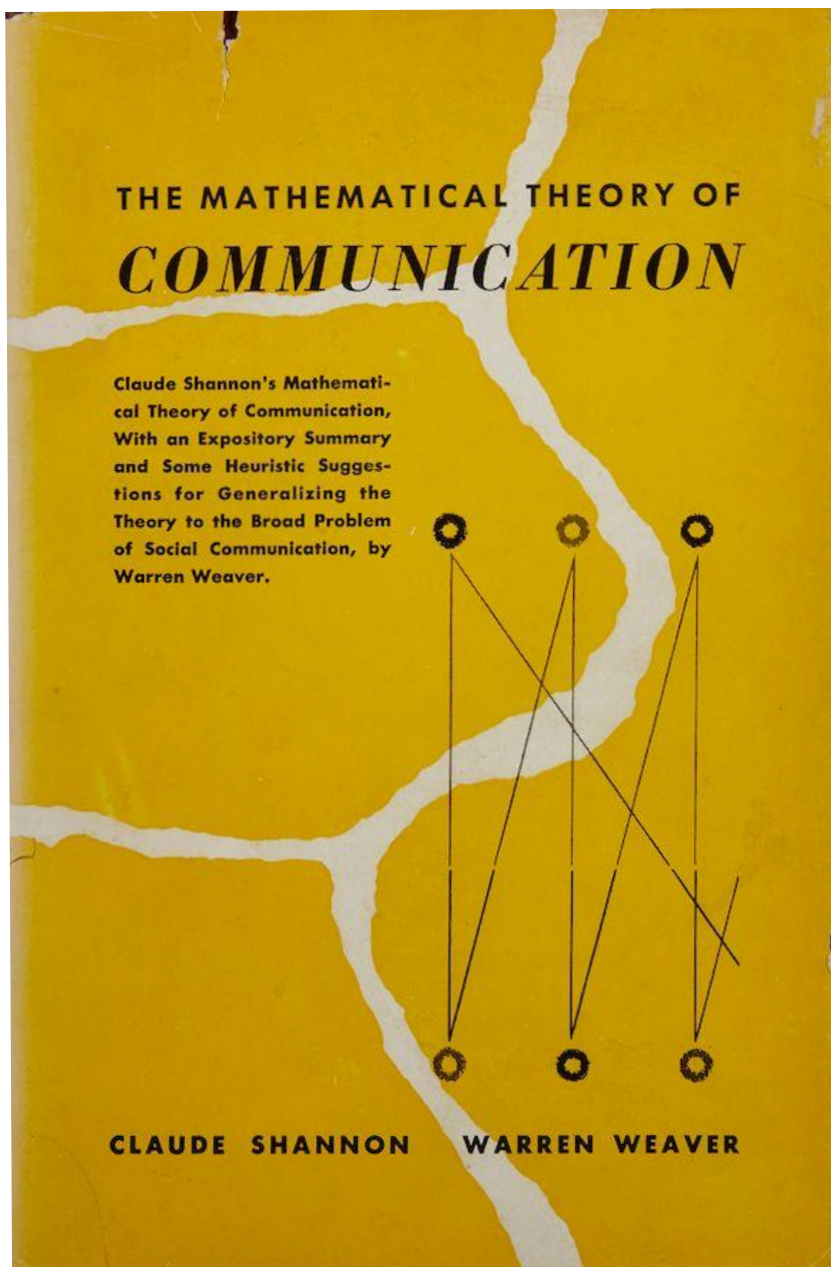


Texts are polysemic

Human language follows a cognitive economy

- it is not possible to generate a new word for each new concept created by the human mind
- the same word can be used for many purposes
- the meaning of each word is disambiguated by the listener with the help of the rich context

De Saussure: the relationship between sound, word and meaning is arbitrary



Texts are ambiguous

The human language is not a code in the Shannon and Weaver meaning

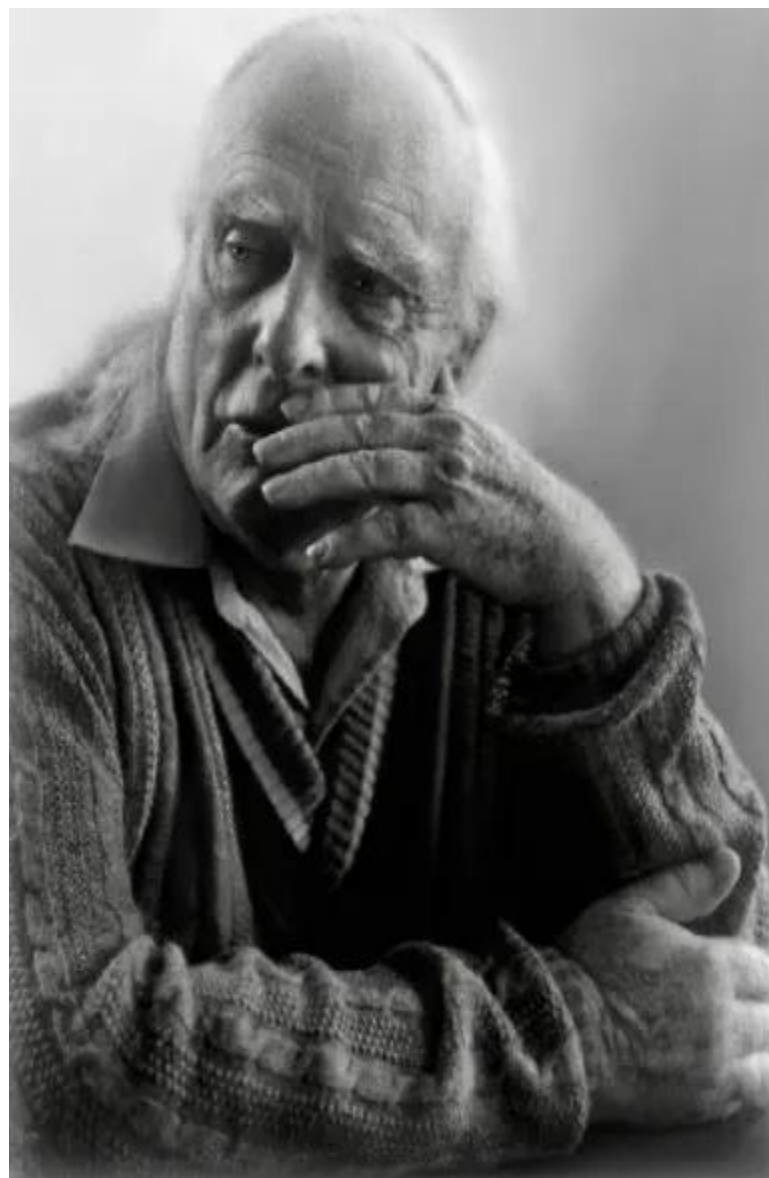
- intended message
- communication channel
- noise
- maximization of the signal/noise ratio

Texts are ambiguous/2

The way in which humans understand each other is by sharing a context

Part of the work of assigning a meaning to a text is carried out by a rich context

e.g. telephone line game



STUDIES

IN · THE

WAY · OF

WORDS

PAUL GRICE

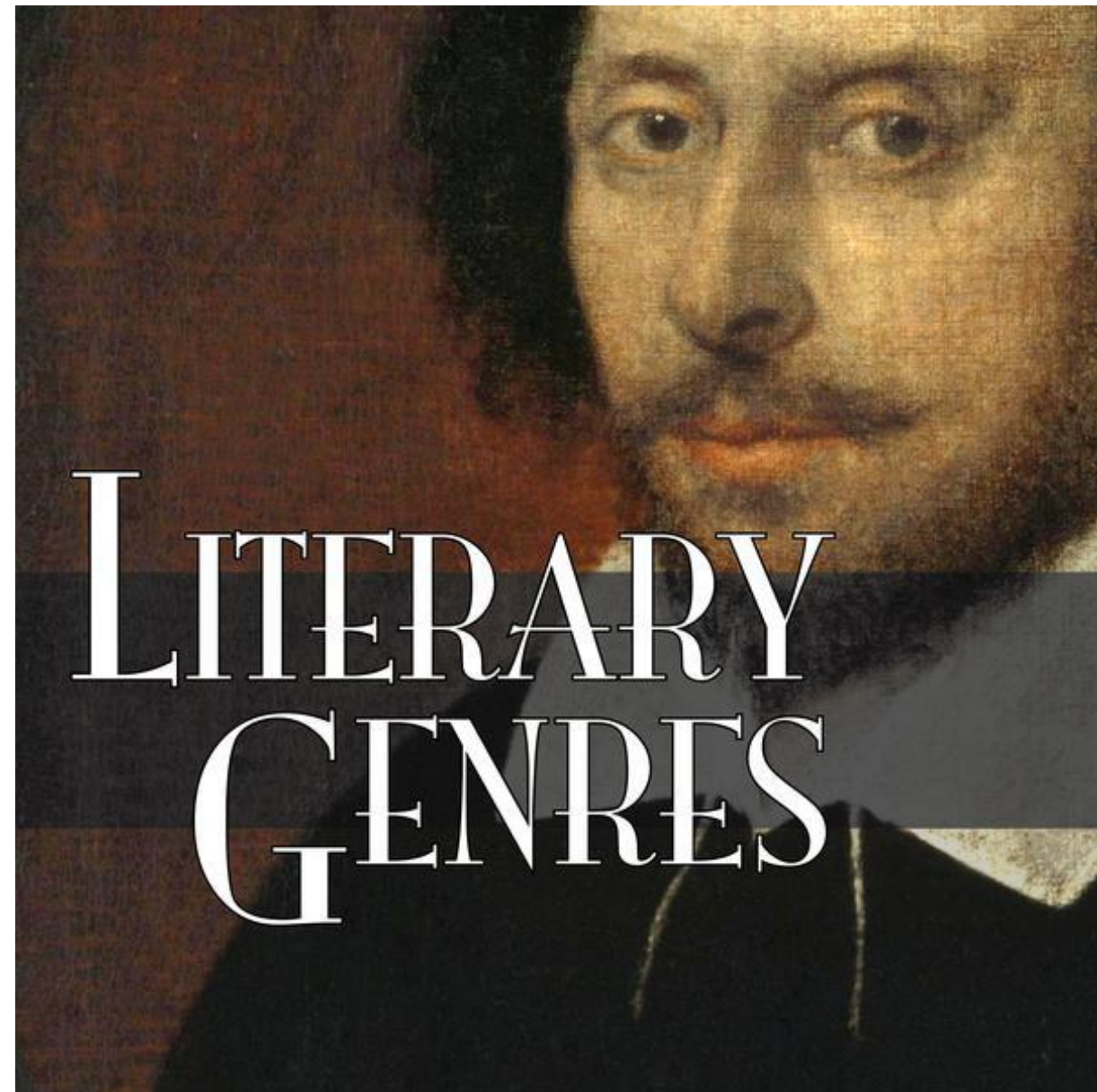
Texts are heterogeneous

In literature they are classified according to an established classification

- prose
 - drama
 - poetry
- etc.

What about non literary texts?

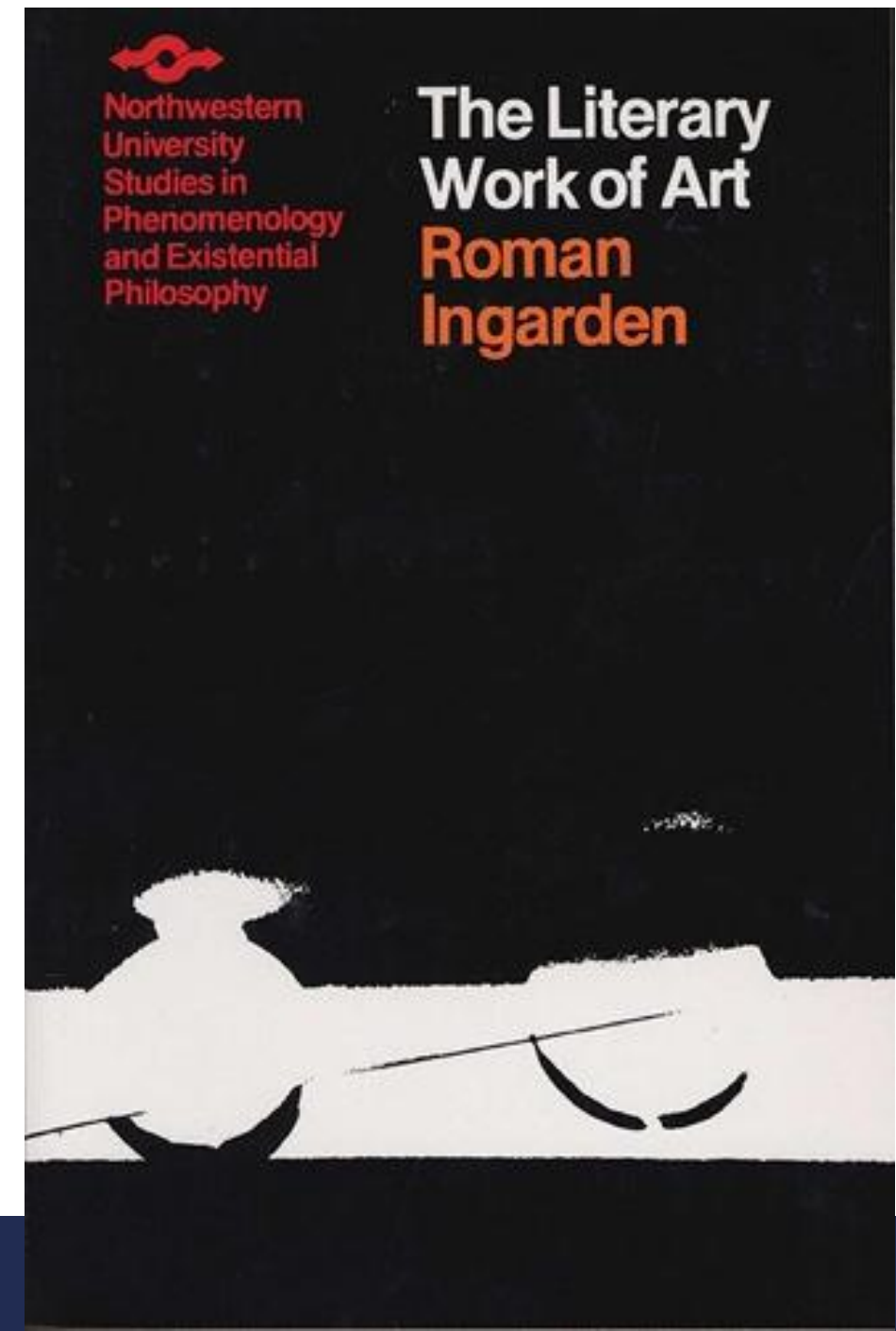
Understanding the style or the genre of a text is necessary to address the meaning



Texts are organized hierarchically

The construction of meaning takes place

- bottom up (from the sentence to the paragraph up to the section and the overall text)
- but also top down- it is at the end of a text that the overall meaning is understood

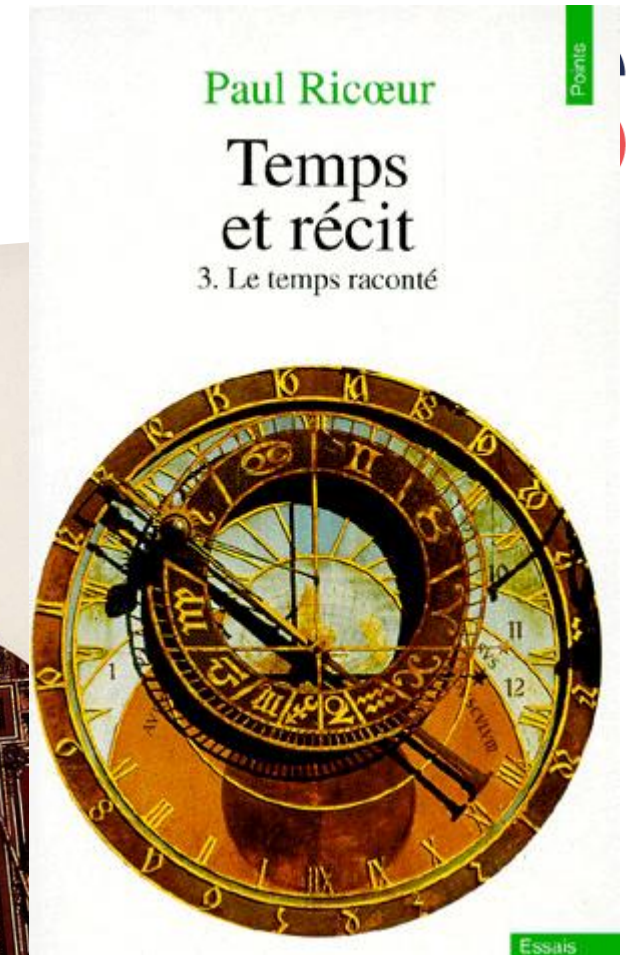
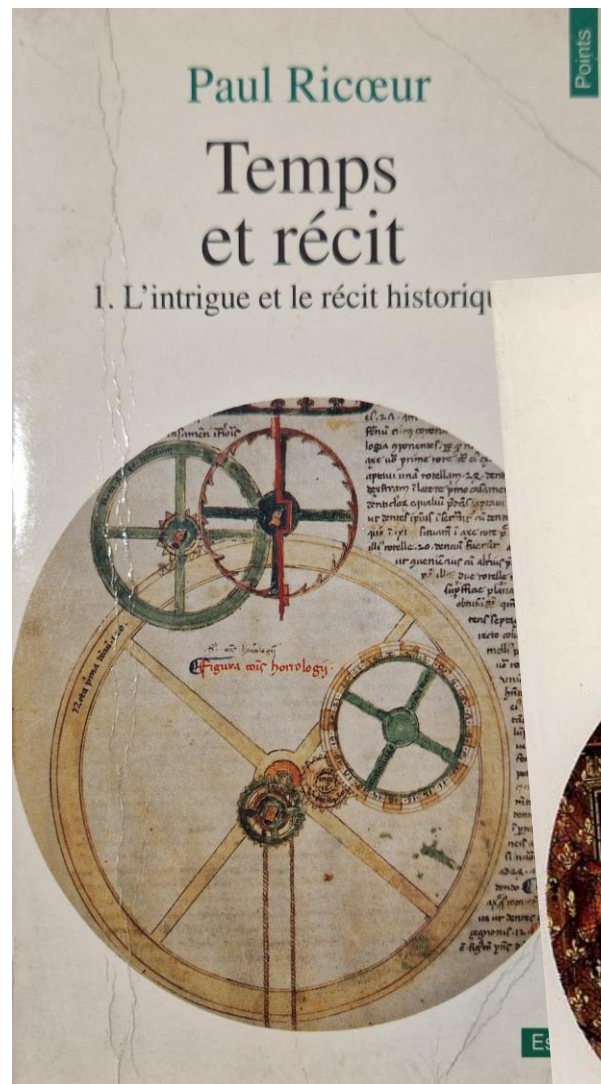


Texts are narrative

The meaning of a text is constructed narratively

- a sequence of events
- embedded in historical / personal time
- complex (plot, intrigue)
- multi-actor

Inference of causal relations via narrative dynamics



Shall we go back home?

For a large class of cases
-although not for all cases-
in which we use the word,
the meaning can be defined as follows:

The meaning of a word
is its use in the language



The day-to-day practice of playing language games recognizes customs and rules.

You shall know a word by the company it keeps !

One of the meanings of “ass” is its habitual collocation with such other words as those above quoted.

John Firth

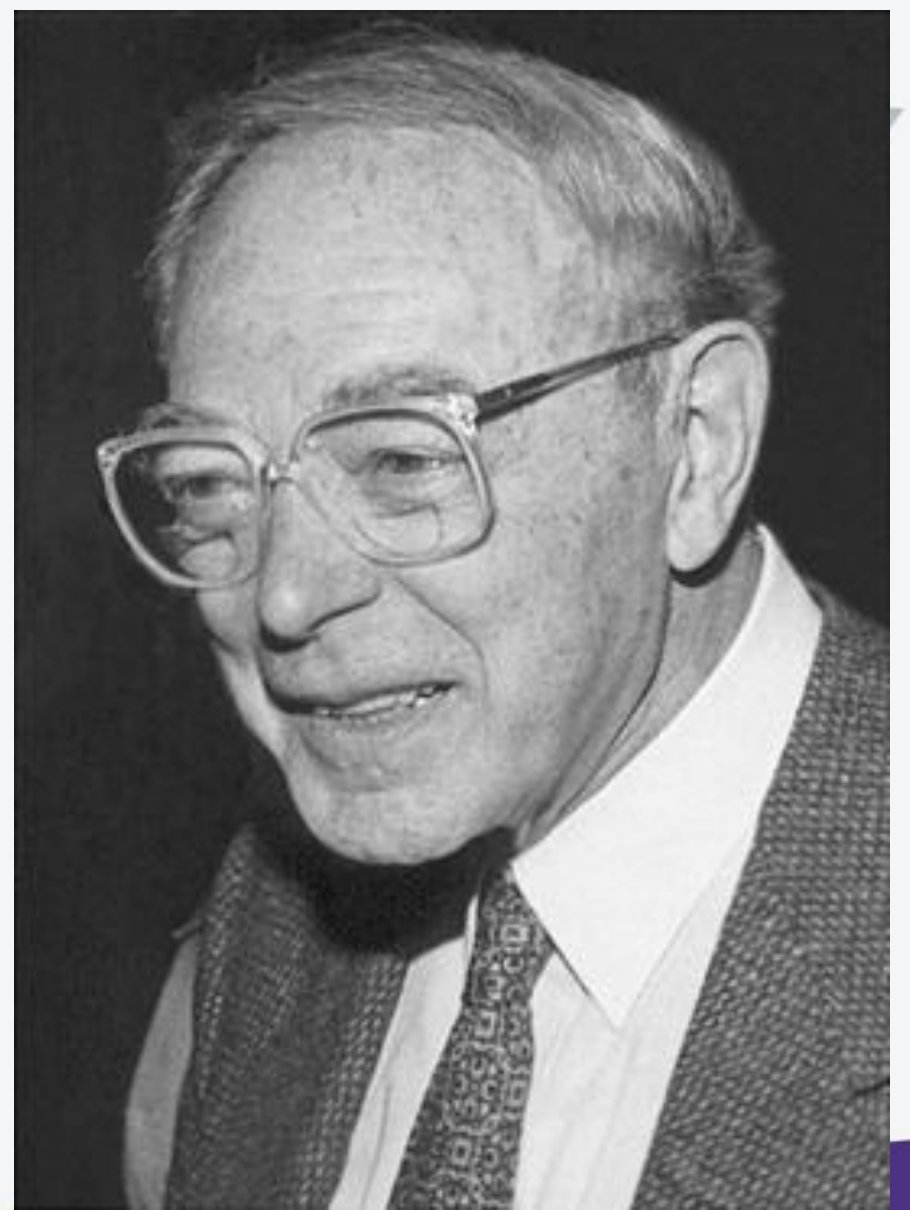


JOHN RUPERT FIRTH

The degree of semantic similarity between two words (or other linguistic entities) can be modeled as a function of the degree of overlapping of respective linguistic contexts

Baroni, M., Lenci, A. (2010)
Distributional memory: A general framework for corpusbased semantics
Computational Linguistics, 36(4), 673-721.

Harris, Zellig. 1954. Distributional structure.
Word, 10(2-3):1456–1162.





Zvi Griliches (1930-1999)

We ourselves do not put enough emphasis on the value of data and data collection in our training of graduate students and in the reward structure of our profession.

We need to teach them to go out and collect their own data on interesting aspects of the economy and to rely less on 'given' data from distant agencies.

Presidential Address

American Economic Association
1993

Innovation economists must build their own datasets.

As a researchers you should know the data one by one.

You must know personally... each cell of your dataset.

Personal communication

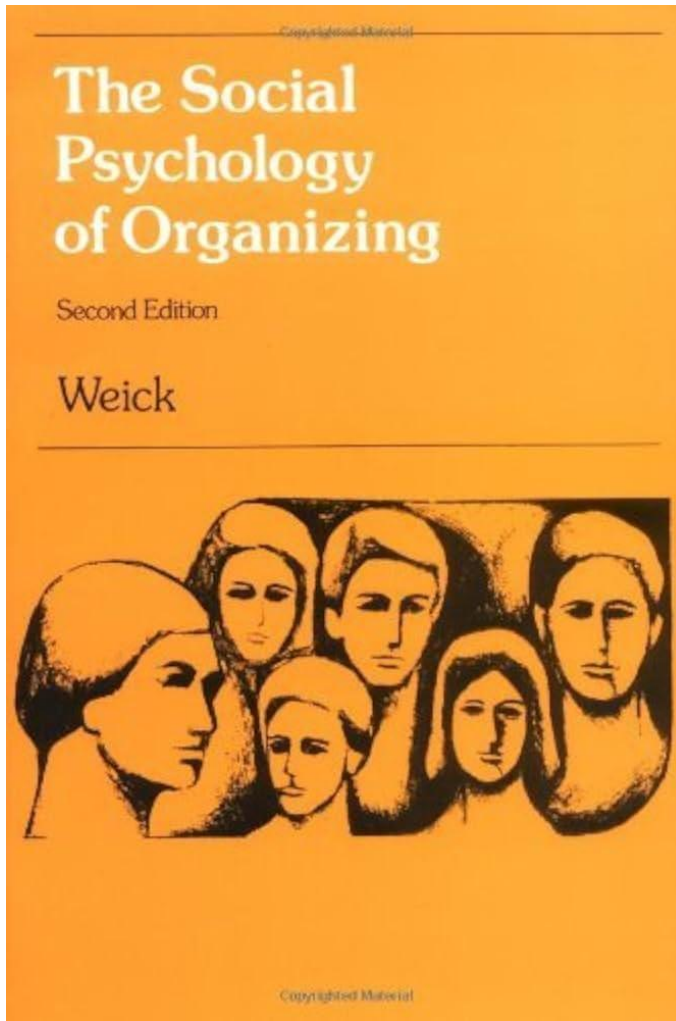


Keith Pavitt (1937-2000)

Case study 1 Strategic management

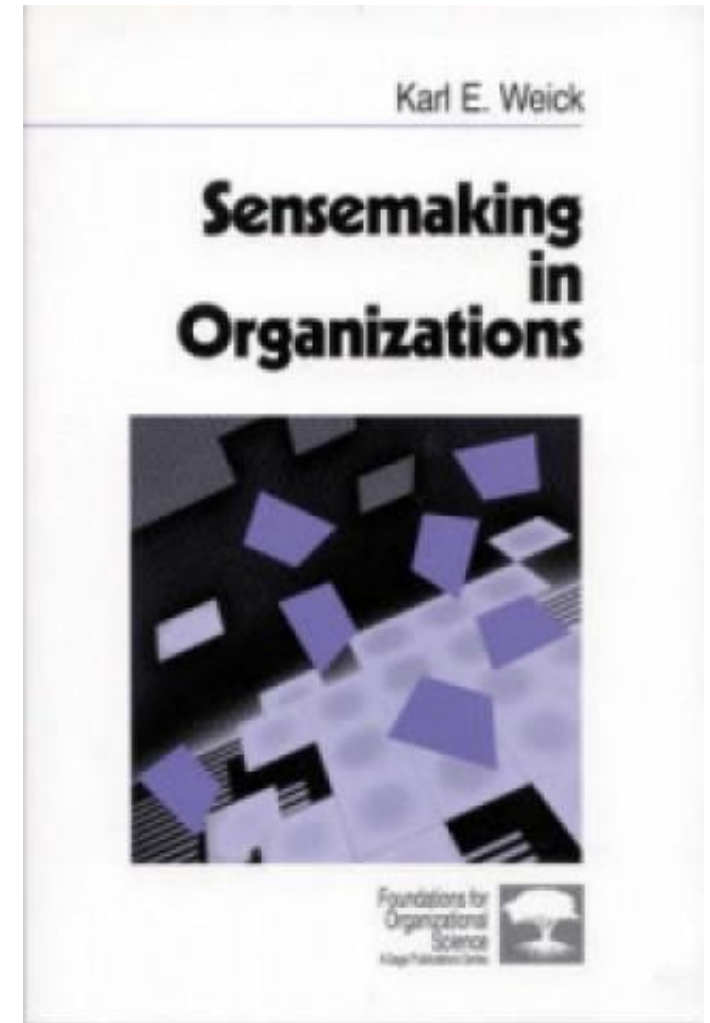
- On-going work (Bonaccorsi et al. 2026)
- Do LLMs produce valid solutions to strategic management problems?
- Strategic management problems
 - Ill-structured (Simon)
 - Wicked (Rittel and Weber)
 - Uncertain imitability (Rumelt)
 - Sense-making (Weick)

Case study 1 Strategic management



Strategy as sensemaking

- enactment
(= reality is not just observed externally but created (co-created) by the strategic actor)
- cues
(= strategic actors do not have access to unambiguous signals but to weak signals, environmental cues, ambiguous and hard-to-interpret information)



Case study 1 Strategic management

- Empirical setting
- Into-the-wild evaluation of LLMs
- Collection of 125 published case studies in business
- Identification (via AI-based summarization) of 40 features that are recurrent
- Experiments: 20 items in input, 20 in (requested) output
- Prompt: solve the business case study by suggesting solutions + Chain-of-Thought

Case study 1 Strategic management

- Main findings
- Solutions proposed by LLMs are all correct and productive
 - but
- Creative ideas are only combinatorial- no abstraction
- Analogies are fluent but limited to semantic regions that are close to the focal firm (e.g. same industry)
- Causal reasoning is flawed by common cause (= same cause for many different effects) and short causal paths
- Counterfactual reasoning is subject to optimistic bias

Case study 2 Creativity



- Discussion of a paper by a top-level author in “science of science” (NBER, 2025)
- Definition of creativity
 - In particular, disruption
- State of the art of the literature on scientific creativity
 - In particular, role of abstraction
- AI and scientific creativity

Definition of creativity



Creativity is clearly a **multidimensional** concept. There is a large debate on which dimensions of creativity are operationalized by existing definitions and measures: the paper does not offer a discussion of alternative approaches.

Atypicality

- Atypical combinations of citations [Uzzi et al. 2013](#); [He, Lei and Wang 2018](#)

Disruption [Funk and Owen-Smith 2017](#)

Why not the following?

Novelty

- First appearance of new words in the text [Arts, Melluso and Veugelers 2024](#)

Discord

- Disconnection [Lin Evans and Wu 2022](#)
- Surprise [Shi and Evans 2019](#); [Foster Shi and Evans 2021](#)

Adjacent possible [Tria et al. 2014](#); [Acemoglu et al. 2016](#); [Monechi et al. 2017](#)

Disruption index

- disruption= a focal paper is highly cited while the references included in it are not cited

But is disruption a clear measure of novelty?

Need to address the criticism about the definition and measure of disruption

Bornmann and Tekles 2019a; 2019b; 2020; Bornmann et al. 2020;

Leydesdorff et al. 2021;

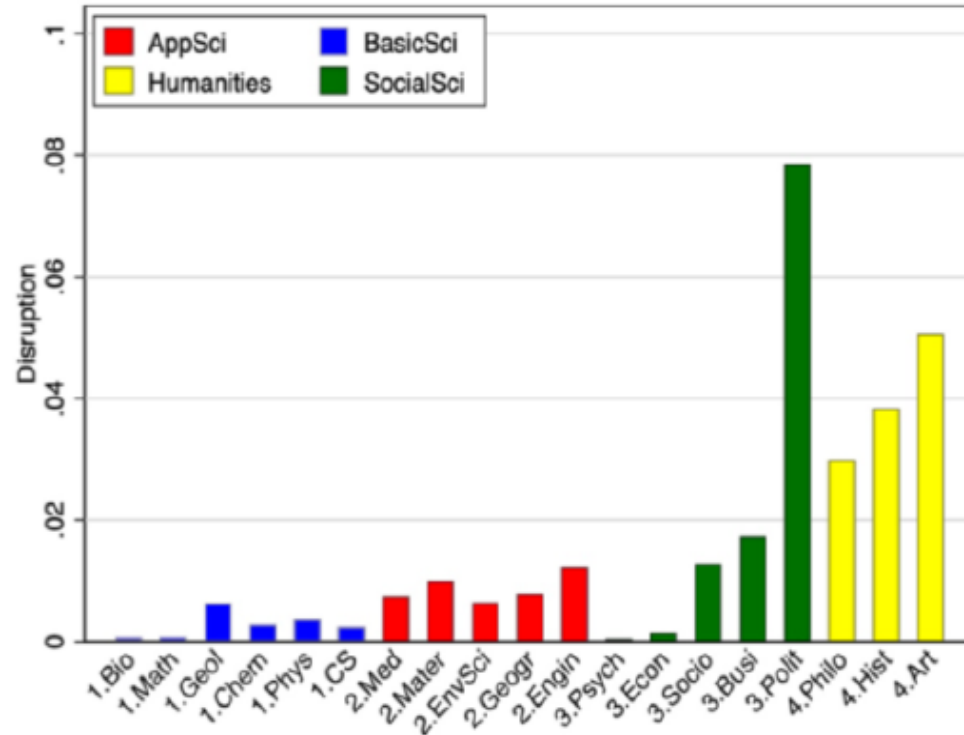
Macher et al. 2024 in *Research Policy*

Noweg et al. 2026 in *Research Policy*

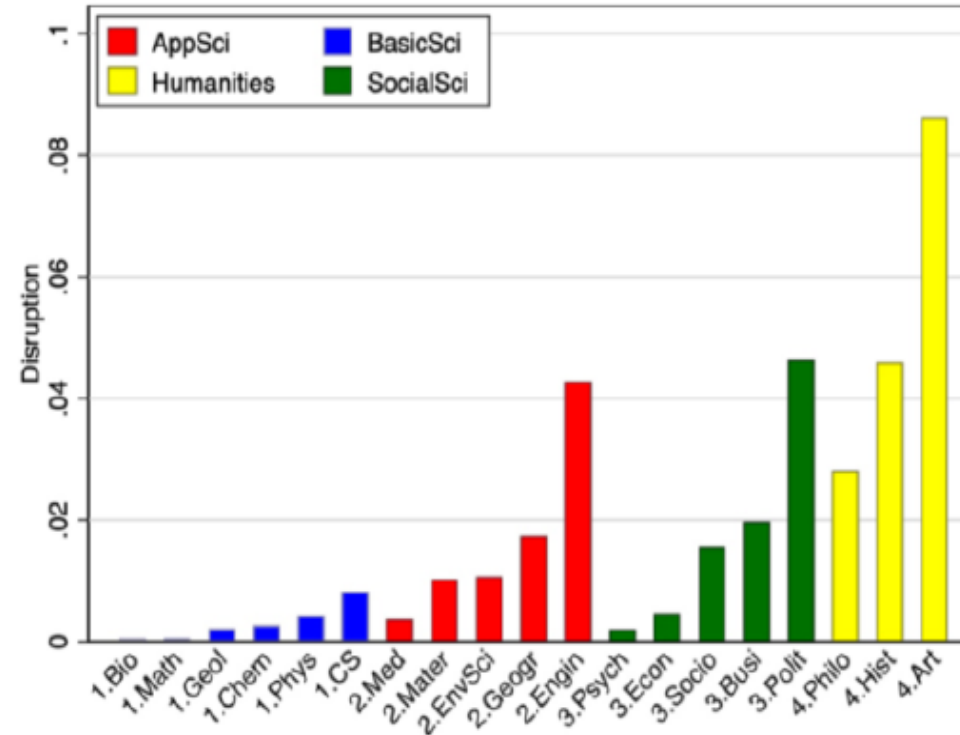
Disruption: why so high in SSH?



Figure 2b. Disruption of AI and Non-AI papers (by Fields)



90th-Percentile Disruption of AI Papers



90th-Percentile Disruption of Non-AI Papers

Theories of creativity



It is not clear what is added by a (rather conventional) survey of the broad literature on creativity

- huge field! Just some classical reference (i.e. Osborn, Amabile, Simonton, Csikszentmihalyi) + product/people/process/place framework

The paper discusses creativity in technology (TRIZ) and science as if they were originated by the same cognitive (*search*) processes. But creativity is domain-specific: different areas of neural circuits are activated by specific creative tasks

[Dunbar 2007](#)

By recruiting this large and broad literature on creativity the paper **creates the illusion to be able to contribute to the field at large.**

Better to focus on scientific creativity only (perhaps reducing the ambitions!)

Theories of scientific creativity



Which **type** of scientific creativity is the paper addressing?

Creativity [Boden 2004](#)

- Combinatorial
- Exploratory
- Transformational

Abduction [Magnani 2000; 2009](#)

- Selective (= inference to the best explanation)
- Creative (=generation of a new plausible hypothesis)

Cognitive perspectives [Thagard 2012; Paul and Kaufman 2014; Schickore 2022](#)

Theories of scientific creativity/2



«**No new fundamental laws of nature are being discovered.** Only a few successful examples have closed the full loop of scientific discovery» [Zenil et al. 2023](#)

«Extracting **human physical insights** from such (AI-based) representations remains challenging» [Iten et al. 2020](#)

«A neural network trained on data from a specific regime may discover regularities that do not generalize in a different regime **whose underlying distribution has shifted**» [Wang et al. 2023](#)

«The large scale automated analysis of the scientific literature, to our knowledge, has **not yet been able to induce new scientific understanding**» [Krenn 2022](#)

Abstraction



A key dimension of creativity in science is *abstraction*, or the ability

- to identify high level concepts (usually *non observable*)
- to manipulate these concepts in models of variable complexity in order to explain observable phenomena.

Abstraction is necessary for

- **Analogical** creativity
- **Counterfactual** thinking

Example from history of science: observation of astronomical phenomena by Copernicus and Keplero by using a model of circular vs ellyptical trajectories (with the same data!)

It is not obvious that abstraction can be achieved via replacement and recombination, which are the operations studied by LCT ([Ding, 2020](#)) and accessible to AI tools.

Abstraction/2



Anyway, there are good reasons to believe that AI (included GenAI) **does not exhibit capabilities in abstract reasoning.**

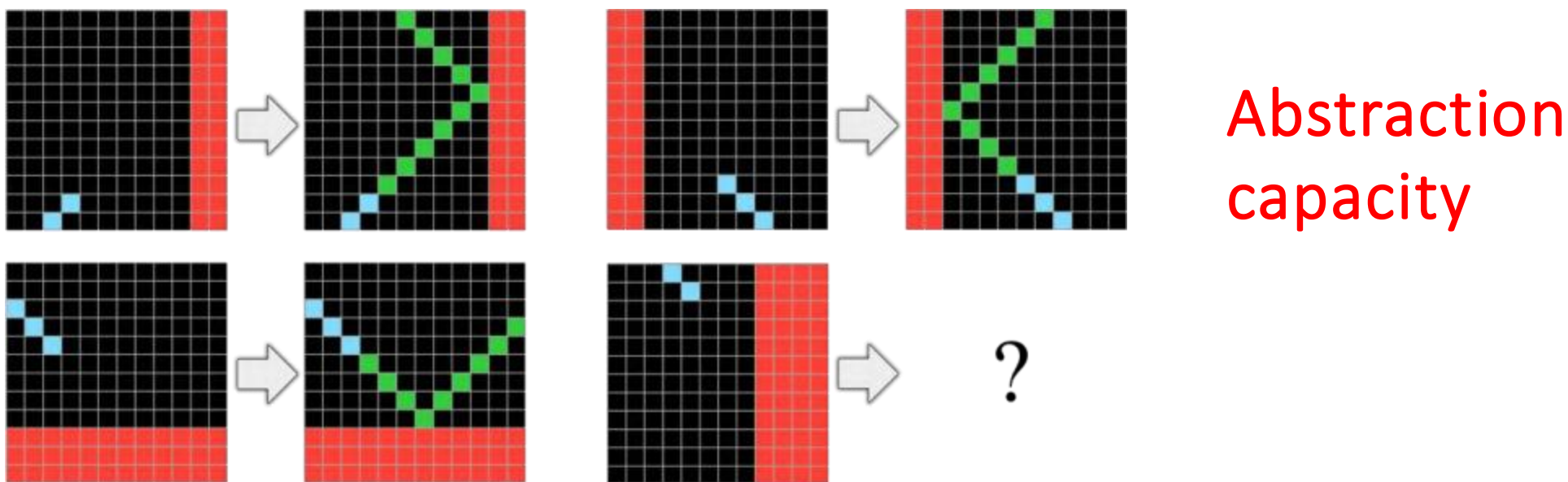
Symbolic AI: **no capacity for generalisation**

Machine learning: robust systems, capable of **only local generalisation**

Generative AI

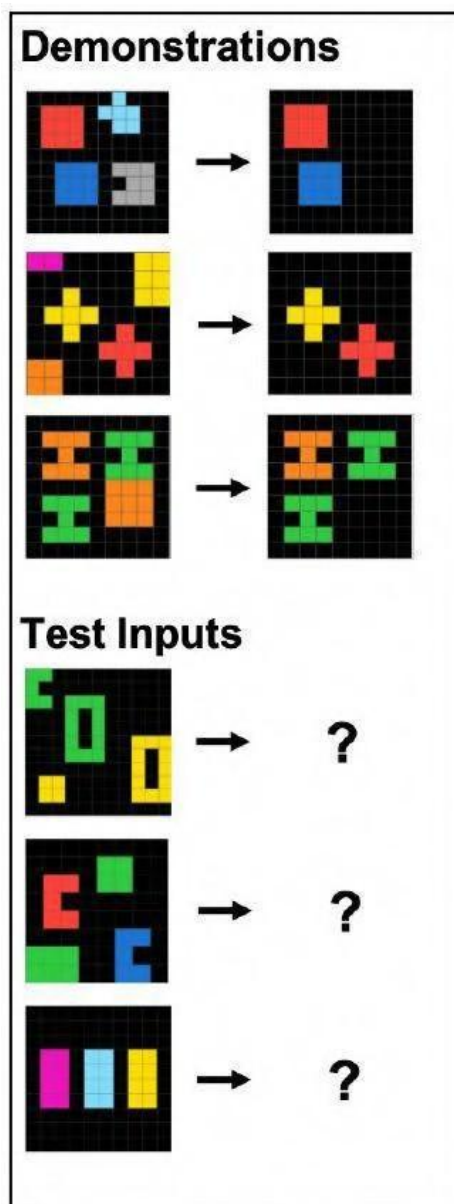
hybrid systems, with various levels of generalisation, capable of handling 'known unknowns', although (to date) **limited capacity to generalise to unknown cases** ('unknown unknowns')

Francois Chollet, Google, Inc. , On the Measure of Intelligence <https://arxiv.org/pdf/1911.01547.pdf>

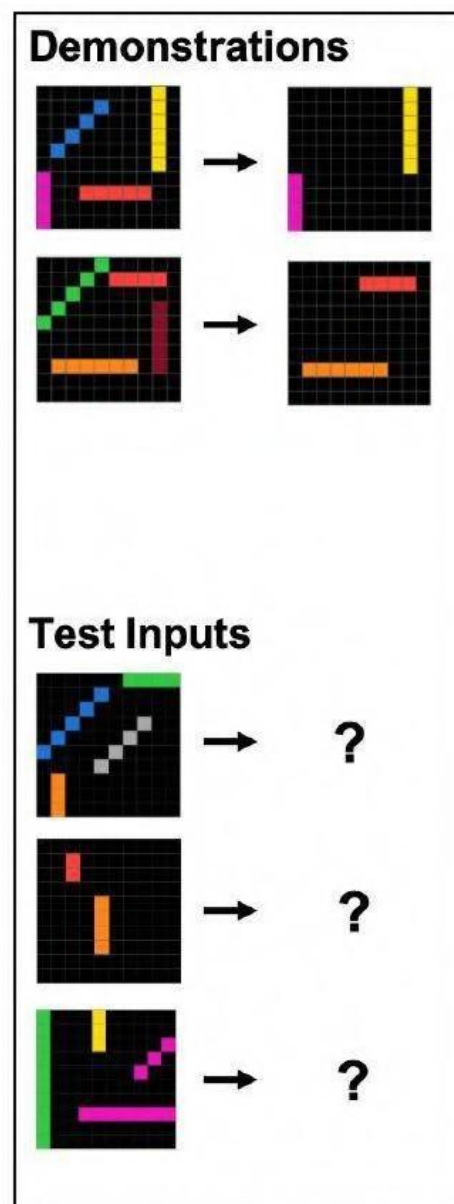


The data show that GPT-4, probably the LLM application with the most general capabilities, is unable to robustly form abstractions and reason about basic concepts in contexts not previously seen in the training data.

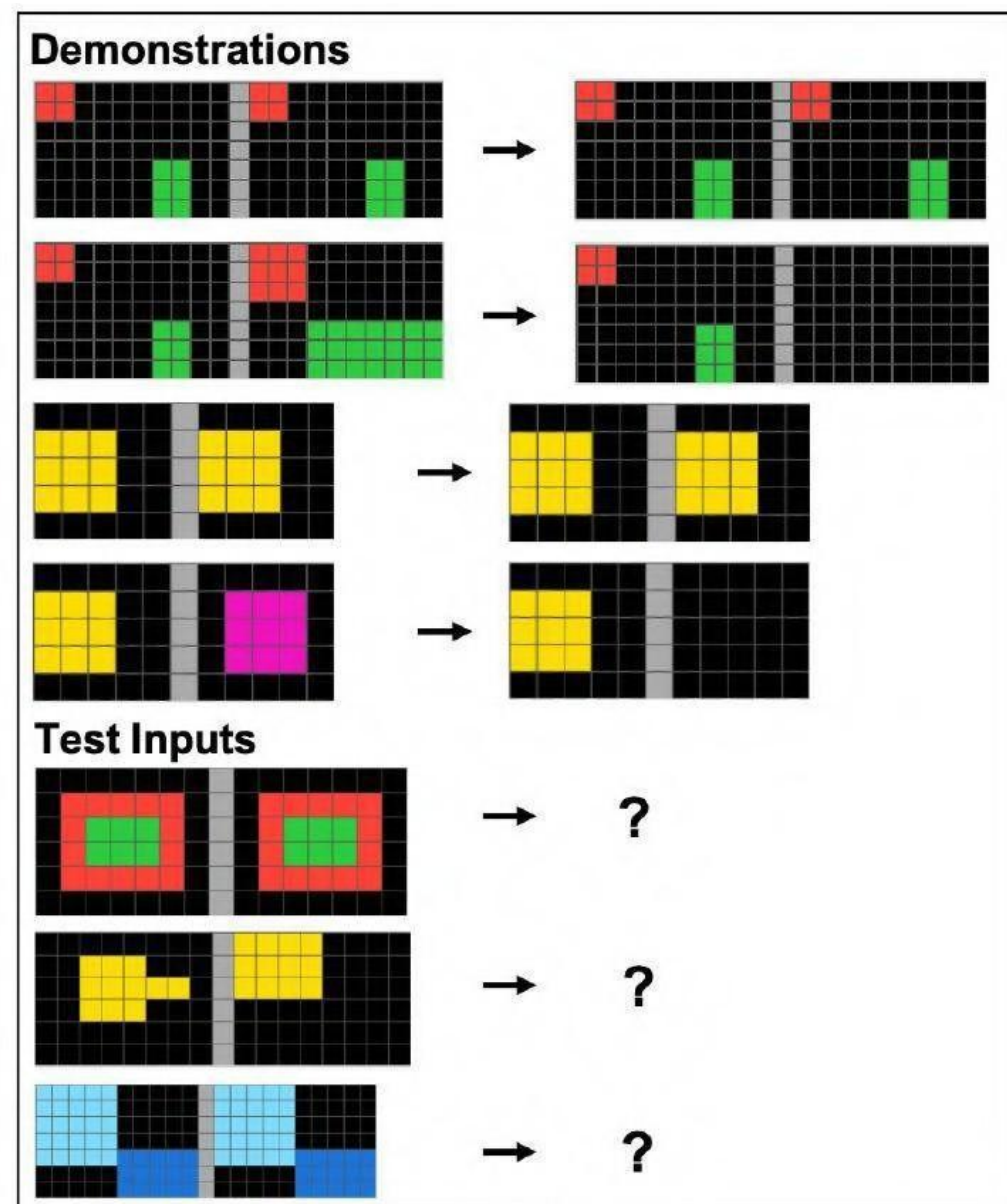
Melanie Mitchell, Alessandro B. Palmarini, and Arseny Moskvichev (2024) Comparing Humans, GPT-4, and GPT-4V On Abstraction and Reasoning Tasks. *arXiv:2311.05247v3*



(a)



(b)



(c)

Scientific understanding



“Scientists would want to comprehend how the oracle made these predictions”.

Scientific understanding

- Empirical adequacy and internal consistency
- Human intelligibility
-
- What is your causal model?

de Regt, H.W. (2020) Understanding, values, and the aims of science. *Philosophy of Science*, 87(5).

Krenn M. (2022) On scientific understanding with artificial intelligence. *Nature Reviews Physics*, 4

Messeri, L., Crockett, M.J. (2024) Artificial intelligence and illusions of understanding in scientific research. *Nature*, 627.

Epistemological fault line	Definition
The Grounding fault	Humans anchor judgment in perceptual, embodied, and social experience, whereas LLMs begin from text alone, reconstructing meaning indirectly from symbols.
The Parsing fault	Humans parse situations through integrated perceptual and conceptual processes; LLMs perform mechanical tokenization that yields a structurally convenient but semantically thin representation.
The Experience fault	Humans rely on episodic memory, intuitive physics and psychology, and learned concepts; LLMs rely solely on statistical associations encoded in embeddings.
The Motivation fault	Human judgment is guided by emotions, goals, values, and evolutionarily shaped motivations; LLMs have no intrinsic preferences, aims, or affective significance.
The Causality fault	Humans reason using causal models, counterfactuals, and principled evaluation; LLMs integrate textual context without constructing causal explanations, depending instead on surface correlations.
The Metacognitive fault	Humans monitor uncertainty, detect errors, and can suspend judgment; LLMs lack metacognition and must always produce an output, making hallucinations structurally unavoidable.
The Value fault	Human judgments reflect identity, morality, and real-world stakes; LLM “judgments” are probabilistic next-token predictions without intrinsic valuation or accountability.

TABLE I. Seven epistemological fault lines marking structural divergences between human and LLM judgment.

Source;
Quattrociocchi,
Capraro and Perc
(2025)

[nature](#) > [news features](#) > [article](#)

NEWS FEATURE | 01 September 2026

Bland new world: is AI making us all think the same?

Researchers worry that generative AI is homogenizing culture and cognition.

By [Matthew Hutson](#)



Epistemological work

The knowledge of humanities (2026)

- four disciplines in humanities
 - History
 - Literary criticism
 - History of art
 - History of architecture
- first hand examination of scientific principles and methods in classical works in these disciplines
- analytical philosophy (of art, history, literature)
- recent advances in epistemology
(causality, counterfactual, history-driven hard science)

PDF freely available at

<https://www.brepols.net/products/978-2-503-61874-6>



Thank you for your attention

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